TUTORIAL 10

Determinants, Adjugates, Similarity, and an Introduction to Diagonalization

MOTIVATION AND GUIDING IDEA

A change of basis preserves the linear map but can greatly simplify its matrix. The determinant tests invertibility without computing an inverse, detects dependence among rows, and measures changes in area or volume. Before a long calculation, look for zeros, linear dependence, or block structure.

NOTATION AND CONVENTIONS

- $M_n(\mathbb{F})$ is the space of square matrices of size n over $\mathbb{F}$; I_n is the identity matrix, and A^T is the transpose.
- $A \sim B$ means that $B = P^{-1}AP$ for some invertible P. The columns of P are the new basis vectors expressed in the old basis.
- $\det A$ is the determinant. The alternative notation $|A|$ sometimes means the same scalar; it is not an absolute value of a matrix.
- M_{ij} is obtained by deleting row i and column j; $C_{ij} = (-1)^{i+j} \det(M_{ij})$ is the corresponding cofactor.
- $R_i \leftarrow R_i + cR_j$ means replacing row i by itself plus c times row j. The determinant preservation rule requires $i \neq j$.

DEFINITION 10.1

For every $n \geq 1$, the determinant is the unique function $\det : M_n(\mathbb{F}) \to \mathbb{F}$ that is linear in each row separately with the other rows fixed, vanishes when two rows are equal, and satisfies $\det I_n = 1$. Row operation rules and cofactor expansions follow. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij}C_{ij}$; the result is independent of the selected row. With the convention $\det(\varnothing) = 1$ for the empty matrix of size 0, the formula also holds for $n = 1$.

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NOTE

For square matrices over any field, $\det(AB) = \det A \det B$, $\det(A^T) = \det A$, and $\det(cA) = c^n \det A$. Also, A is invertible exactly when $\det A \neq 0$; then $\det(A^{-1}) = (\det A)^{-1}$. The determinant is not generally additive: $\det(A+B) \neq \det A + \det B$ in general.

NOTE

Distinguish two transformations. Similarity, $B = P^{-1}AP$, describes an operator after the same basis change in domain and codomain; it preserves the determinant. Congruence, $B = P^T AP$, describes a basis change for a bilinear form; here $\det B = (\det P)^2 \det A$. The formulas cannot be interchanged.

NOTE

The geometric interpretation requires a real matrix. In $\mathbb{R}^2$, $|\det A|$ is the area scale factor; in $\mathbb{R}^3$, it is the volume scale factor. If $\det A = 0$, the image has smaller dimension. If $\det A \neq 0$, the sign indicates whether orientation is preserved or reversed.

10.1 Further Determinant Expansions and Properties

10.1.1 The Determinant Formula

RECALL AND TOOLS

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a square matrix of size n. For $1 \leq i, j \leq n$, the **minor matrix** corresponding to entry (i, j), denoted M_{ij}, is the square matrix of size $n-1$ obtained by deleting row i and column j from A.

EXAMPLE 10.1

Let $A = \begin{pmatrix} 4 & -1 & 3 \\ 2 & 5 & 0 \\ -2 & 1 & 6 \end{pmatrix}$. Then $M_{11} = \begin{pmatrix} 5 & 0 \\ 1 & 6 \end{pmatrix}$ and $M_{23} = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$.

DEFINITION 10.2

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be square of size n. Its **determinant**, denoted $\det(A)$, can be computed recursively in n:

1. If $n = 1$, then $A = (a_{11})$ and we define
$$\det(A) = a_{11}.$$
2. If $n = 2$, then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$

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3. If $n \geq 3$, then for every $1 \leq i \leq n$, expansion along row i gives

$$\det(A) = \sum_{j=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}),$$

where M_{ij} is obtained by deleting row i and column j.

Equivalently, for every $1 \leq j \leq n$, expansion along column j gives

$$\det(A) = \sum_{i=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}).$$

EXERCISE 10.1

Please compute the determinants again, this time using cofactor expansion and its consequences.

1. For $t \in \mathbb{R}$,

$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 9 \end{pmatrix}.$$

2.

$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -2 & -6 & 6 & -4 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Expand along the first column. Its two nonzero entries are in positions $(1,1)$ and $(4,1)$, so

$$\det A(t) = \det M_{11} - 2 \det M_{41}.$$

The minor matrices are

$$M_{11} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & t & 9 \end{pmatrix}, \qquad M_{41} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Expand the first along its first column and the second along its first row:

$$\det M_{11} = (1 \cdot 9 - 2t) + (2 \cdot 2 - 1 \cdot 1) = 12 - 2t,$$

$$\det M_{41} = 1(2 \cdot 2 - 1 \cdot 1) + (-1)(1 \cdot 1 - 2 \cdot 0) = 3 - 1 = 2.$$

Thus $\det A(t) = 12 - 2t - 2 \cdot 2 = 8 - 2t$, as obtained by the previous method.

2. Regard the determinant as a function of the rows. Since $r_3 = r_1 - 3r_2$, linearity in the

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EXERCISE 10.1 CONTINUED

third row gives

$$\det(r_1, r_2, r_1 - 3r_2, r_4, r_5) = \det(r_1, r_2, r_1, r_4, r_5) \\ - 3\det(r_1, r_2, r_2, r_4, r_5) = 0.$$

Each term has two equal rows. Linearity in a row and vanishing for equal rows follow from the determinant expansion, so again $\det B = 0$.

EXERCISE 10.2

Please find all $k \in \mathbb{R}$ for which

$$A(k) = \begin{pmatrix} 2 & k & 4 \\ 3 & k+1 & 6 \\ -1 & -1 & k+3 \end{pmatrix}$$

is singular, using its determinant.

WORKED SOLUTION

Expanding along the first row gives

$$\begin{aligned} \det A(k) &= 2\big((k+1)(k+3)+6\big) - k\big(3(k+3)+6\big) + 4(k-2) \\ &= 2k^2 + 8k + 18 - 3k^2 - 15k + 4k - 8 \\ &= -k^2 - 3k + 10 = -(k+5)(k-2). \end{aligned}$$

Thus the matrix is singular exactly when $k = -5$ or $k = 2$.

EXERCISE 10.3

1. Let $n \geq 2$ and let $x_1, \ldots, x_n$ be real (or complex) numbers. Define the Vandermonde matrix

$$V(x_1, \ldots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Please prove that

$$\det\big(V(x_1, \ldots, x_n)\big) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

WORKED SOLUTION

1. Write

$$\Delta_n(x_1, \ldots, x_n) = \det\big(V(x_1, \ldots, x_n)\big).$$

We prove by induction on n that

$$\Delta_n(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

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EXERCISE 10.3 CONTINUED

Base case, $n = 2$:

$$\Delta_2(x_1, x_2) = \det \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1,$$

as required.

Inductive step: assume the statement holds for $n-1$, and prove it for n. Regard Δ_n as a polynomial in x_n, with $x_1, \ldots, x_{n-1}$ fixed. Expanding the determinant along row n gives

$$\Delta_n(x_1, \ldots, x_n) = \sum_{j=1}^{n} x_n^{j-1} \, (-1)^{n+j} \, \det(M_{nj}),$$

Thus Δ_n is a polynomial in x_n of degree at most $n-1$. The only term containing x_n^{n-1} comes from column n, so its coefficient is

$$(-1)^{n+n} \det(M_{nn}) = \det\big(V(x_1, \ldots, x_{n-1})\big) = \Delta_{n-1}(x_1, \ldots, x_{n-1}).$$

If two of $x_1, \ldots, x_{n-1}$ are equal, the matrix has two equal rows, and both sides of the desired formula are zero. Hence we may assume these numbers are pairwise distinct. For every $k \in \{1, \ldots, n-1\}$, substituting $x_n = x_k$ makes rows k and n of V equal, so

$$\Delta_n(x_1, \ldots, x_{n-1}, x_k) = 0.$$

Therefore $(x_n - x_k)$ divides Δ_n for every $k = 1, \ldots, n-1$ (as a polynomial in x_n). Since these roots are distinct and the degree is at most $n-1$, there is a polynomial P such that

$$\Delta_n(x_1, \ldots, x_n) = \Big(\prod_{k=1}^{n-1} (x_n - x_k)\Big) \, P(x_1, \ldots, x_{n-1}),$$

where P is independent of x_n.

Compare the coefficients of x_n^{n-1} on both sides. On the left the coefficient is $\Delta_{n-1}(x_1, \ldots, x_{n-1})$; on the right the coefficient of x_n^{n-1} in $\prod_{k=1}^{n-1}(x_n - x_k)$ is 1. Hence

$$P(x_1, \ldots, x_{n-1}) = \Delta_{n-1}(x_1, \ldots, x_{n-1}).$$

It follows that

$$\Delta_n(x_1, \ldots, x_n) = \Delta_{n-1}(x_1, \ldots, x_{n-1}) \prod_{k=1}^{n-1} (x_n - x_k).$$

Finally, by the induction hypothesis,

$$\Delta_{n-1}(x_1, \ldots, x_{n-1}) = \prod_{1 \le i < j \le n-1} (x_j - x_i),$$

and therefore

$$\Delta_n(x_1, \ldots, x_n) = \Big(\prod_{1 \le i < j \le n-1} (x_j - x_i)\Big)\Big(\prod_{k=1}^{n-1} (x_n - x_k)\Big) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

This proves the formula.

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EXERCISE 10.4

1. Let $n \geq 2$ and let $w \in \mathbb{C}$ be a primitive nth root of unity: $w^n = 1$ and $w^k \neq 1$ for every $1 \leq k \leq n-1$. Define the matrix

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} \in M_n(\mathbb{C}).$$

Please compute $\det(F)$ explicitly.

2. Let $n \geq 3$, let $1 \leq p \leq n-1$, and please write $q = n - p$. For $\alpha, \beta \in \mathbb{R}$, define $A(\alpha, \beta) \in M_n(\mathbb{R})$ by

$$a_{ii} = 1 \quad (1 \leq i \leq n),$$

and, for every $i \neq j$,

$$a_{ij} = \begin{cases} \alpha, & 1 \leq i, j \leq p \text{ or } p+1 \leq i, j \leq n, \\ \beta, & \text{otherwise.} \end{cases}$$

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $T(v) = A(\alpha, \beta)v$. Please determine for which α, β the map T is an isomorphism, and please compute $\det(A(\alpha, \beta))$.

WORKED SOLUTION

First display the structure of the matrices:

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}.$$

$$A(\alpha, \beta) = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right), .$$

The block sizes are

$$A_{11} \in M_p(\mathbb{R}), \qquad A_{22} \in M_q(\mathbb{R}),$$
$$A_{12} \in M_{p\times q}(\mathbb{R}), \qquad A_{21} \in M_{q\times p}(\mathbb{R}).$$

$$A_{11} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{p\times p}, \qquad A_{22} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{q\times q}.$$

$$A_{12} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{p\times q}, \qquad A_{21} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{q\times p}.$$

$$P^{-1}A(\alpha, \beta)P = (1-\alpha)I_{n-2} \oplus B = \left(\begin{array}{c|c} (1-\alpha)I_{n-2} & 0 \\ \hline 0 & B \end{array}\right),$$

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EXERCISE 10.4 CONTINUED

where the lower-right block has size 2:

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

1. **Fact used below (Vandermonde determinant).** For $x_1, \dots, x_n \in \mathbb{C}$, define

$$V(x_1, \dots, x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n}.$$

Then

$$\det\big(V(x_1, \dots, x_n)\big) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Set

$$x_i = w^{i-1} \qquad (i = 1, \dots, n).$$

For every i, j,

$$F_{ij} = w^{(i-1)(j-1)} = (w^{i-1})^{j-1} = x_i^{j-1}.$$

Therefore

$$F = V(x_1, \dots, x_n).$$

By the preceding exercise on the Vandermonde determinant,

$$\det(F) = \prod_{1 \le i < j \le n} (x_j - x_i) = \prod_{0 \le i < j \le n-1} \left(w^j - w^i\right).$$

For every $0 \le i < j \le n-1$,

$$w^j - w^i = w^i\left(w^{j-i} - 1\right),$$

so

$$\det(F) = \left(\prod_{0 \le i < j \le n-1} w^i \right) \left(\prod_{0 \le i < j \le n-1} (w^{j-i} - 1) \right).$$

Write

$$S = \sum_{0 \le i < j \le n-1} i.$$

Then

$$\prod_{0 \le i < j \le n-1} w^i = w^S.$$

We now compute S: For each $i \in \{0, \dots, n-2\}$, there are exactly $(n-1-i)$ values of j with $i < j \le n-1$, so

$$S = \sum_{i=0}^{n-2} i(n-1-i) = \frac{n(n-1)(n-2)}{6}.$$

Consequently,

$$\prod_{0 \le i < j \le n-1} w^i = w^{\frac{n(n-1)(n-2)}{6}}.$$

Also group the factors by $m = j - i \in \{1, \dots, n-1\}$. For each m, there are exactly

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EXERCISE 10.4 CONTINUED

$(n-m)$ pairs (i,j) with $j-i=m$, so

$$\prod_{0\le i<j\le n-1}(w^{j-i}-1)=\prod_{m=1}^{n-1}(w^m-1)^{n-m}.$$

Therefore

$$\boxed{\det(F)=w^{\frac{n(n-1)(n-2)}{6}}\prod_{m=1}^{n-1}(w^m-1)^{n-m}}.$$

Because w is a primitive nth root of unity, $w^m\neq 1$ for $1\le m\le n-1$, so $w^m-1\neq 0$. Every factor in the product is therefore nonzero, and $\det(F)\neq 0$.

2. **Theorems used below (from class).**

(a) Isomorphisms and representing matrices. Let V be a finite-dimensional real vector space, let $T:V\to V$ be linear, and let $[T]_{\mathcal{B}}$ represent T in a basis $\mathcal{B}$. Then

$$T \text{ is an isomorphism } \iff [T]_{\mathcal{B}} \text{ is invertible.}$$

(b) Invertibility and the determinant. For every $A\in M_n(\mathbb{R})$,

$$A \text{ is invertible } \iff \det(A)\neq 0.$$

(This is the standard determinant criterion for invertibility.)

(c) Invariance of the determinant under change of basis. If P is invertible, then

$$\det(P^{-1}AP)=\det(A),$$

because

$$\det(P^{-1}AP)=\det(P^{-1})\det(A)\det(P)=\det(A).$$

We can now solve the problem.

Write $A=A(\alpha,\beta)$. Define

$$U=\Big\{x\in\mathbb{R}^n : x_{p+1}=\cdots=x_n=0,\quad \sum_{i=1}^{p}x_i=0\Big\},$$

$$W=\Big\{x\in\mathbb{R}^n : x_1=\cdots=x_p=0,\quad \sum_{i=p+1}^{n}x_i=0\Big\}.$$

Clearly U,W are subspaces, since they are defined by homogeneous linear equations in the coordinates.

We first prove that $Ax=(1-\alpha)x$ for every $x\in U$. Let $x\in U$. Then $x_{p+1}=\cdots=x_n=0$ and $\sum_{j=1}^{p}x_j=0$.

If $1\le i\le p$, the structure of row i of A (diagonal entry 1, entries α elsewhere in its block, and entries β between blocks) gives

$$(Ax)_i=1\cdot x_i+\sum_{\substack{1\le j\le p\\ j\neq i}}\alpha x_j+\sum_{j=p+1}^{n}\beta x_j.$$

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EXERCISE 10.4 CONTINUED

The last sum vanishes because $x_{p+1} = \cdots = x_n = 0$, so

$$(Ax)_i = x_i + \alpha \sum_{\substack{1 \le j \le p \\ j \ne i}} x_j.$$

Now

$$\sum_{\substack{1 \le j \le p \\ j \ne i}} x_j = \sum_{j=1}^{p} x_j - x_i = 0 - x_i = -x_i,$$

and hence

$$(Ax)_i = x_i + \alpha(-x_i) = (1-\alpha)x_i.$$

If $p+1 \le i \le n$, the coefficients of coordinates $1, \ldots, p$ in this row are all β (the second block of x is zero). Thus

$$(Ax)_i = \sum_{j=1}^{p} \beta x_j = \beta \sum_{j=1}^{p} x_j = 0 = (1-\alpha)x_i.$$

Therefore $Ax = (1-\alpha)x \in U$. In particular, U is A-invariant, and the restriction $A|_U$ is multiplication by $(1-\alpha)$. The same argument gives $Ay = (1-\alpha)y$ for every $y \in W$. Thus W is also A-invariant and $A|_W = (1-\alpha)\mathrm{Id}_W$.

Define

$$u = (\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}), \qquad v = (\underbrace{0, \ldots, 0}_{p}, \underbrace{1, \ldots, 1}_{q}), \qquad S = \mathrm{span}\{u, v\}.$$

We next compute Au and Av.

For every $1 \le i \le p$,

$$(Au)_i = 1 \cdot 1 + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha \cdot 1 + \sum_{j=p+1}^{n} \beta \cdot 0 = 1 + (p-1)\alpha.$$

and, for every $p+1 \le i \le n$,

$$(Au)_i = \sum_{j=1}^{p} \beta \cdot 1 + \Big(1 \cdot 0 + \sum_{\substack{p+1 \le j \le n \\ j \ne i}} \alpha \cdot 0\Big) = p\beta.$$

Therefore

$$Au = \big(1 + (p-1)\alpha\big)u + p\beta\, v.$$

Similarly,

$$Av = q\beta\, u + \big(1 + (q-1)\alpha\big)v.$$

Thus S is A-invariant, and its representing matrix in the basis (u, v) is

$$B = \begin{pmatrix} 1 + (p-1)\alpha & q\beta \\ p\beta & 1 + (q-1)\alpha \end{pmatrix}.$$

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EXERCISE 10.4 CONTINUED

We claim that

$$\mathbb{R}^n = (U \oplus W) \oplus S.$$

First, $U \cap W = \{0\}$, since a vector in both subspaces must have every coordinate zero. Also, if $au + bv \in U \oplus W$, the sum of its first-block coordinates is $ap = 0$, so $a = 0$; the sum of its second-block coordinates is $bq = 0$, so $b = 0$. Hence $S \cap (U \oplus W) = \{0\}$.

Next, every $x \in \mathbb{R}^n$ admits a decomposition $x = x_{UW} + x_S$ with $x_{UW} \in U \oplus W$ and $x_S \in S$. Write

$$s_1 = \sum_{i=1}^{p} x_i, \qquad s_2 = \sum_{i=p+1}^{n} x_i,$$

and define

$$x_S = \frac{s_1}{p}u + \frac{s_2}{q}v \in S, \qquad x_{UW} = x - x_S.$$

The sum of the first-block coordinates of x_{UW} is $s_1 - s_1 = 0$, and the second-block sum is $s_2 - s_2 = 0$, so $x_{UW} \in U \oplus W$. This proves the direct-sum decomposition.

Choose a basis $\mathcal{B}$ by concatenating bases of U and W and the basis (u, v) of S. In this basis, the matrix of A has the block diagonal form

$$(1-\alpha)I_{n-2} \oplus B.$$

Let P be the change-of-coordinates matrix from $\mathcal{B}$ to the standard basis. Then

$$P^{-1}AP = (1-\alpha)I_{n-2} \oplus B.$$

Combining facts (b) and (c),

$$\det(A) = \det(P^{-1}AP) = \det\big((1-\alpha)I_{n-2} \oplus B\big).$$

The determinant of a block diagonal matrix is the product of the determinants of its blocks (this follows by cofactor expansion, or by a suitable factorization and multiplicativity $\det(AB) = \det(A)\det(B)$). Therefore

$$\det(A) = \det\big((1-\alpha)I_{n-2}\big) \cdot \det(B) = (1-\alpha)^{n-2}\det(B).$$

Compute $\det(B)$:

$$\det(B) = \big(1 + (p-1)\alpha\big)\big(1 + (q-1)\alpha\big) - pq\,\beta^2.$$

and hence

$$\boxed{\det(A) = (1-\alpha)^{n-2}\Big(\big(1 + (p-1)\alpha\big)\big(1 + (q-1)\alpha\big) - pq\,\beta^2\Big)}.$$

By fact (a), T is an isomorphism if and only if its representing matrix A in the standard basis is invertible. By fact (b), this is equivalent to $\det(A) \neq 0$. Thus

$$T \text{ is an isomorphism} \iff \det(A) \neq 0 \iff (1-\alpha)^{n-2} \neq 0 \text{ and } \det(B) \neq 0.$$

Since $\alpha, \beta \in \mathbb{R}$ and $n \geq 3$, $(1-\alpha)^{n-2} \neq 0$ holds if and only if $\alpha \neq 1$. The exact condition

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EXERCISE 10.4 CONTINUED

is therefore

$$\boxed{\alpha \neq 1} \quad \text{and} \quad \boxed{\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) \neq pq\,\beta^2}.$$

EXERCISE 10.5

For $s \in \mathbb{R}$, define $T_s : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T_s(v) = A_s v, \qquad A_s = \begin{pmatrix} 3 & 1 \\ s & 2 \end{pmatrix}.$$

The triangle Δ has vertices $(0,0)$, $(1,0)$, $(0,1)$. Please find the vertices and area of its image, please determine when T_s is invertible, and please describe the image when $s = 6$.

WORKED SOLUTION

The images of the vertices are $(0,0)$, $(3,s)$, $(1,2)$. The original triangle has area $1/2$, and the area scale factor is $|\det A_s| = |3\cdot 2 - 1\cdot s| = |6-s|$. Hence the image has area $|6-s|/2$. Also, T_s is invertible exactly when $6 - s \neq 0$, namely when $s \neq 6$. When $s = 6$, the columns satisfy $(3,6)^T = 3(1,2)^T$. Every image is therefore a multiple of $(1,2)^T$, and the triangle collapses to a line segment on $y = 2x$, with zero area. For $s < 6$, orientation is preserved; for $s > 6$, it is reversed. For example, at $s = 1$ the image has area $5/2$.

Additional practice

EXERCISE 10.6 EXAM QUESTION

1. Let $A \in M_n(\mathbb{R})$ satisfy $\operatorname{rank}(A^2) < \operatorname{rank}(A)$. Please prove that there exists $v \in \mathbb{R}^n$ such that $A^2 v = 0$ and $Av \neq 0$.

2. Let $A, B \in M_n(\mathbb{R})$. Please prove that

$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq n + \operatorname{rank}(AB).$$

   Please deduce that if $\operatorname{rank}(A) + \operatorname{rank}(B) > n$, then $AB \neq 0$.

3. Let $N \in M_n(\mathbb{R})$ satisfy $N^2 = 0$ and $\operatorname{rank}(N) = r$. Please find $\det(I_n + tN)$ and $\operatorname{rank}(I_n + tN)$ for every $t \in \mathbb{R}$.

WORKED SOLUTION

1. If $Av = 0$, then $A^2 v = A(Av) = 0$. Hence $\ker A \subseteq \ker A^2$. By the rank-nullity theorem and the given inequality,

$$\begin{aligned} \dim\ker A^2 &= n - \operatorname{rank}(A^2) \\ &> n - \operatorname{rank}(A) = \dim\ker A. \end{aligned}$$

   Thus the inclusion is strict: $\ker A \subsetneq \ker A^2$. Choose $v \in \ker A^2 \setminus \ker A$. Membership in the first kernel gives $A^2 v = 0$, while exclusion from the second gives $Av \neq 0$. This vector remains nonzero after one application of A and becomes zero after two applications.

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EXERCISE 10.6 EXAM QUESTION CONTINUED

2. Consider the restricted map

$$T : \operatorname{Im} B \longrightarrow \mathbb{R}^n, \qquad T(u) = Au.$$

Its image is exactly $\operatorname{Im}(AB)$: every $u \in \operatorname{Im} B$ has the form $u = Bv$, so $T(u) = ABv$; conversely, every vector ABv arises in this way. Its kernel is $\operatorname{Im} B \cap \ker A$. Applying rank-nullity to T gives

$$\operatorname{rank}(B) = \dim(\operatorname{Im} B \cap \ker A) + \operatorname{rank}(AB).$$

However,

$$\dim(\operatorname{Im} B \cap \ker A) \leq \dim \ker A = n - \operatorname{rank}(A).$$

Substitute and rearrange:

$$\begin{aligned} \operatorname{rank}(B) &\leq n - \operatorname{rank}(A) + \operatorname{rank}(AB), \\ \operatorname{rank}(A) + \operatorname{rank}(B) &\leq n + \operatorname{rank}(AB). \end{aligned}$$

If the ranks sum to more than n, then $\operatorname{rank}(AB) > 0$, so AB is not the zero matrix.

3. First compute the two products:

$$\begin{aligned} (I_n + tN)(I_n - tN) &= I_n - tN + tN - t^2N^2 = I_n, \\ (I_n - tN)(I_n + tN) &= I_n + tN - tN - t^2N^2 = I_n. \end{aligned}$$

Thus $I_n + tN$ is invertible for every $t \in \mathbb{R}$, with inverse $I_n - tN$. Immediately,

$$\operatorname{rank}(I_n + tN) = n.$$

To compute the determinant, note that $N^2 = 0$ implies $\operatorname{Im} N \subseteq \ker N$. Hence $r \leq n - r$, or $2r \leq n$. Choose a basis $(u_1, \ldots, u_r)$ of $\operatorname{Im} N$ and extend it to a basis $(u_1, \ldots, u_r, w_1, \ldots, w_{n-2r})$ of $\ker N$. For each j, choose v_j with $Nv_j = u_j$.

The list

$$\mathcal{B} = (u_1, \ldots, u_r, w_1, \ldots, w_{n-2r}, v_1, \ldots, v_r)$$

is a basis. Indeed, applying N to a vanishing linear combination gives a combination of the u_j, so every coefficient of a v_j is zero. The remaining coefficients vanish because they multiply a basis of $\ker N$. Thus the list contains n linearly independent vectors. In this basis,

$$[N]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & I_r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad [I_n + tN]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 & tI_r \\ 0 & I_{n-2r} & 0 \\ 0 & 0 & I_r \end{pmatrix}.$$

A block of size zero is simply omitted. This is upper triangular with every diagonal entry equal to 1. Since the determinant is preserved under change of basis,

$$\det(I_n + tN) = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Neither answer depends on r: the determinant is 1 and the rank is n.

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EXERCISE 10.7 FOUNDATION

For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 3 & 6 & t \end{pmatrix}.$$

1. Please compute $\det A_t$ using one row operation, and please determine when A_t is invertible.
2. At $t = 0$, please find a basis of $\ker A_0$ and please find $\operatorname{rank} A_0$.
3. Please construct B_t from A_t by, in this order, interchanging the first two rows, multiplying the third row by 3, and adding 5 times the third row to the first row. Please compute $\det B_t$ without writing B_t explicitly.

WORKED SOLUTION

1. The operation $R_3 \leftarrow R_3 - 3R_1$ preserves the determinant and gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & t \end{pmatrix}.$$

   Hence $\det A_t = 1 \cdot 4 \cdot t = 4t$, and A_t is invertible exactly when $t \neq 0$.

2. At $t = 0$, the kernel equations are $x_1 + 2x_2 = 0$ and $4x_2 + x_3 = 0$. Setting $x_2 = s$ gives

$$\ker A_0 = \{s(-2, 1, -4)^T : s \in \mathbb{R}\} = \operatorname{span}\{(-2, 1, -4)^T\}.$$

   The kernel has dimension 1, so rank-nullity gives $\operatorname{rank} A_0 = 3 - 1 = 2$.

3. The interchange multiplies the determinant by -1, scaling the row multiplies it by 3, and the final operation preserves it. Thus

$$\det B_t = (-1) \cdot 3 \cdot \det A_t = -12t.$$

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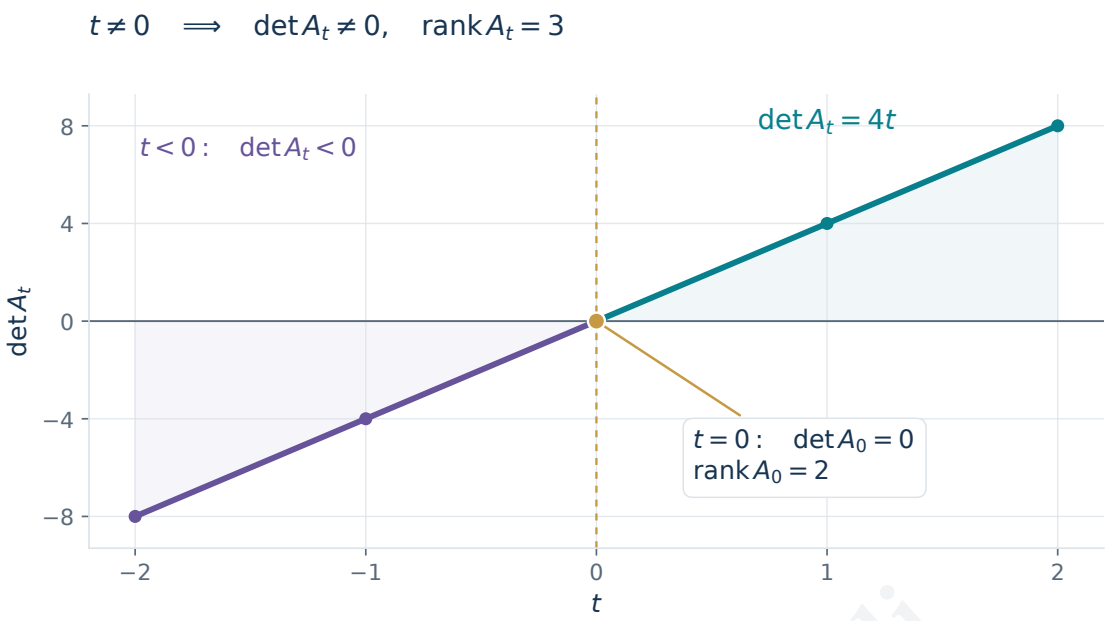


Figure 10.1. For the matrix A_t in the exercise, the graph $\det A_t = 4t$ crosses zero only at $t = 0$. For every $t \neq 0$, the matrix is invertible and has rank 3; at the gold point, $\operatorname{rank} A_0 = 2$, with kernel $\operatorname{span}\{(-2, 1, -4)^T\}$.

EXERCISE 10.8 CHALLENGE

Let $A \in M_n(\mathbb{F})$ be invertible, $U \in \mathbb{F}^{n\times k}$, and $V \in \mathbb{F}^{k\times n}$, with $n, k \geq 1$.

1. Please use determinants of block matrices to prove

$$\det(A + UV) = \det A \, \det(I_k + VA^{-1}U).$$

   Do not assume that U or V is square or invertible.

2. For $s \in \mathbb{R}$, please use the identity to compute the determinant of

$$C_s = \begin{pmatrix} s+2 & 0 & 1 \\ 0 & s+2 & -1 \\ s+1 & s+1 & 1 \end{pmatrix}$$

   and please determine when it is invertible. At the singular parameter value, please find its kernel and rank.

WORKED SOLUTION

1. First prove $\det(I_n + XY) = \det(I_k + YX)$ for $X \in \mathbb{F}^{n\times k}$ and $Y \in \mathbb{F}^{k\times n}$. Set

$$K = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}, \quad L = \begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix}, \quad H = \begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix}.$$

   Both L and H are block triangular with determinant 1. We have

$$LK = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}, \qquad HK = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

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EXERCISE 10.8 CHALLENGE CONTINUED

Multiplicativity and the determinant formula for block triangular matrices give

$$\det K = \det(I_k + YX) = \det(I_n + XY).$$

Now $A + UV = A(I_n + A^{-1}UV)$. Substitute $X = A^{-1}U$ and $Y = V$ to obtain the required identity.

2. Take $A = I_3$ and write $C_s = I_3 + UV$, where

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} s+1 & 0 & 1 \\ 0 & s+1 & -1 \end{pmatrix}.$$

Direct multiplication verifies this factorization and gives

$$I_2 + VU = \begin{pmatrix} s+3 & 1 \\ -1 & s+1 \end{pmatrix}.$$

Therefore

$$\det C_s = \det(I_2 + VU) = (s+1)(s+3) + 1 = (s+2)^2.$$

Thus C_s is invertible exactly when $s \neq -2$. At $s = -2$, its kernel equations are $x_3 = 0$ and $-x_1 - x_2 + x_3 = 0$, so

$$\ker C_{-2} = \operatorname{span}\{(1, -1, 0)^T\}, \qquad \operatorname{rank} C_{-2} = 3 - 1 = 2.$$

The multiplicity of a root of the determinant as a polynomial in s is not the kernel dimension: this root is double, while the kernel has dimension 1.

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EXERCISE 10.9 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be invertible and satisfy $AB = -BA$. Please prove that n is even. For every $m \geq 1$, please construct such a pair of size $2m$ with $A^2 = B^2 = I_{2m}$.

WORKED SOLUTION

Determinant rules give

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero, so cancellation gives $1 = (-1)^n$. Over $\mathbb{R}$, n must be even. For every $m \geq 1$, take

$$A = \begin{pmatrix} I_m & 0 \\ 0 & -I_m \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}.$$

Direct block multiplication gives

$$AB = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} = -AB,$$

and $A^2 = B^2 = I_{2m}$. Thus both are invertible.

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EXERCISE 10.10 CHALLENGE

Let $n \geq 1$, and let $x_1, \ldots, x_n$ be distinct elements of a field F, and let $y_1, \ldots, y_n$ be distinct. Assume $x_i + y_j \neq 0$ for every i, j, and define

$$C = \left(\frac{1}{x_i + y_j} \right)_{i,j=1}^{n}.$$

Please prove that every square submatrix of C is invertible. Please deduce that the real matrix $H_n = (1/(i+j-1))_{i,j=1}^{n}$ is invertible for every $n \geq 1$.

WORKED SOLUTION

First suppose $Cc = 0$, and define

$$q(t) = \sum_{j=1}^{n} c_j \prod_{k \neq j} (t + y_k).$$

This polynomial has degree at most $n-1$. Row i of the equation gives

$$q(x_i) = \left(\prod_{k=1}^{n} (x_i + y_k) \right) \sum_{j=1}^{n} \frac{c_j}{x_i + y_j} = 0.$$

It has n distinct roots and must therefore be the zero polynomial. At $t = -y_j$, every term except term j vanishes, so

$$0 = q(-y_j) = c_j \prod_{k \neq j} (y_k - y_j).$$

The product is nonzero, forcing $c_j = 0$ for every j. Thus C is invertible. A square submatrix selects equally many row and column indices. The selected elements remain distinct within each list, and their sums are still nonzero, so the same proof applies. For the final assertion, take $x_i = i - 1$ and $y_j = j$ over $\mathbb{R}$. Every denominator $i + j - 1$ is positive, and $C = H_n$.

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10.2 The Classical Adjugate

DEFINITION 10.3

Let $A = (a_{ij}) \in M_n(\mathbb{F})$ with $n \geq 2$. Write $A_{\widehat{i,j}}$ for the matrix obtained by deleting row i and column j. The scalar

$$C_{ij} = (-1)^{i+j} \det A_{\widehat{i,j}}$$

is its (i, j) cofactor. The classical adjugate is

$$\operatorname{adj}(A) = (C_{ij})^T.$$

For $n = 1$, set $\operatorname{adj}((a)) = (1)$. The notation $\operatorname{adj}(A)$ denotes the classical adjugate. The Hermitian adjoint will be denoted by $A^* = \overline{A}^T$.

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PROPOSITION 10.1

Every square matrix over a field satisfies

$$A\,\mathrm{adj}(A) = \mathrm{adj}(A)A = (\det A)I_n.$$

Indeed, entry (i, j) of the first product is

$$(A\,\mathrm{adj}(A))_{ij} = \sum_{k=1}^{n} a_{ik}C_{jk}.$$

For $i = j$, this is expansion along row j. For $i \neq j$, it is the determinant expansion of the matrix obtained by replacing row j with row i. That matrix has two identical rows and determinant zero. Expansion along columns proves the identity for the second product. In particular, if $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A).$$

EXAMPLE 10.2

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$(C_{ij}) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \qquad \mathrm{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Both products can be computed directly:

$$A\,\mathrm{adj}(A) = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I_2,$$

$$\mathrm{adj}(A)A = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = (ad - bc)I_2.$$

When $ad - bc \neq 0$, division by that scalar gives the usual inverse formula.

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EXERCISE 10.11

Consider the real matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix}.$$

Please compute $\det A$, every cofactor, $\operatorname{adj}(A)$, and A^{-1}. Please verify both adjugate product identities.

WORKED SOLUTION

The matrix is triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2$. The minors and signs give

$$\begin{aligned} C_{11} &= 1 \cdot 2 - 3 \cdot 0 = 2, & C_{12} &= -(0 \cdot 2 - 3 \cdot 0) = 0, & C_{13} &= 0 \cdot 0 - 1 \cdot 0 = 0, \\ C_{21} &= -(2 \cdot 2 - 0 \cdot 0) = -4, & C_{22} &= 1 \cdot 2 - 0 \cdot 0 = 2, & C_{23} &= -(1 \cdot 0 - 2 \cdot 0) = 0, \\ C_{31} &= 2 \cdot 3 - 0 \cdot 1 = 6, & C_{32} &= -(1 \cdot 3 - 0 \cdot 0) = -3, & C_{33} &= 1 \cdot 1 - 2 \cdot 0 = 1. \end{aligned}$$

Thus

$$\operatorname{adj}(A) = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -3/2 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

Direct multiplication verifies

$$A \operatorname{adj}(A) = \begin{pmatrix} 2 & -4+4 & 6-6 \\ 0 & 2 & -3+3 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3,$$

$$\operatorname{adj}(A) A = \begin{pmatrix} 2 & 4-4 & -12+12 \\ 0 & 2 & 6-6 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3.$$

Dividing by 2 also verifies $AA^{-1} = A^{-1}A = I_3$.

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EXERCISE 10.12

Let $A \in M_n(\mathbb{F})$ with $n \geq 2$.

1. Please prove that $\text{adj}(A)$ has rank n, 1, or 0 according as rank $A = n$, rank $A = n-1$, or rank $A \leq n-2$.
2. Please deduce that $\det(\text{adj}(A)) = (\det A)^{n-1}$.

WORKED SOLUTION

1. We use the rank characterization by minors. If a matrix has rank r, select r independent columns. The resulting matrix has row rank r, so selecting r independent rows gives an invertible $r \times r$ minor. Every larger minor has dependent columns and determinant zero. If A is invertible, then $\text{adj}(A) = (\det A)A^{-1}$ is invertible and has rank n. If rank $A = n-1$, some minor of order $n-1$ is nonzero. At least one cofactor is therefore nonzero, so $\text{adj}(A) \neq 0$. On the other hand,

$$A\,\text{adj}(A) = 0,$$

so every column of the adjugate belongs to $\ker A$. Rank-nullity gives

$$\dim \ker A = n - (n-1) = 1.$$

Hence $1 \leq \text{rank}(\text{adj}(A)) \leq 1$, and the rank is 1. If rank $A \leq n-2$, every minor of order $n-1$ vanishes. Every cofactor is zero, so $\text{adj}(A) = 0$.

2. For invertible A, take determinants in $A\,\text{adj}(A) = (\det A)I_n$:

$$\det A \cdot \det(\text{adj}(A)) = (\det A)^n.$$

Division by $\det A \neq 0$ gives the result. If A is singular, part 1 gives $\text{rank}(\text{adj}(A)) \leq 1 < n$, so both sides of the required identity are zero.

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10.3 Similar matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are similar if there is an invertible $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\text{trace}(A) = \text{trace}(B)$$
3. $$\text{rank}(A) = \text{rank}(B)$$

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EXERCISE 10.13

$$A, B \in \mathbb{F}^{n\times n}.$$

Please prove or disprove: if A, B are similar, then A^k, B^k are similar for every $k \geq 1$; if A^k, B^k are similar for some $k \geq 1$, then A, B are similar.

WORKED SOLUTION

The first claim is true: cancelling adjacent inverse factors gives similarity through the same P. The second claim is false over every field. The displayed cubes agree, but a matrix similar to zero must be zero, whereas B is nonzero.

$$B = P^{-1}AP \quad \Longrightarrow \quad B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

$$A = 0_{3\times 3}, \qquad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$A^3 = B^3 = 0, \qquad B^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B \neq 0.$$

EXERCISE 10.14

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \quad T(x, y)^T = (2x + y, x - y)^T, \qquad B = ((1, 1)^T, (1, -2)^T).$$

Please compute $[T]_E^B$, please find v with $T(v) = (9, -6)^T$, and please check the result in both coordinate systems.

WORKED SOLUTION

The columns are the images in standard output coordinates. Dividing by 3 gives the input B-coordinates, and rebuilding in B gives the vector itself.

$$T(1, 1)^T = (3, 0)^T, \qquad T(1, -2)^T = (0, 3)^T, \qquad [T]_E^B = 3I_2.$$

$$[v]_B = (3, -2)^T, \qquad v = 3(1, 1)^T - 2(1, -2)^T = (1, 7)^T.$$

$$T(1, 7)^T = (2 + 7, 1 - 7)^T = (9, -6)^T, \qquad 3I_2(3, -2)^T = (9, -6)^T.$$

10.4 Similar Matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are called similar if there is an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

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1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 10.15

Let $A, B \in \mathbb{F}^{n\times n}$. Please prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation using $PP^{-1} = I$ gives
$$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$
Thus the powers are similar via the same matrix P.
2. The statement is false. Over any field, take
$$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$
We have $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P, whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, the example $A = -2I_2$, $B = 2I_2$ also works: their squares are equal, and a scalar matrix is similar only to itself. This example gives no counterexample in characteristic 2, where $-2I_2 = 2I_2$.

Common pitfall: The equality $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking a power is not an injective operation on matrices, as these examples show.

10.5 Similar Matrices

DEFINITION 10.4

Let $A, B \in M_n(\mathbb{F})$.
We say that A is **similar** to B (and write $A \sim B$) if there is an invertible $P \in M_n(\mathbb{F})$ such that
$$A = P^{-1}BP$$

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PROPOSITION 10.2

Let $A, B \in M_n(\mathbb{F})$. They are similar if and only if there exist an n-dimensional vector space V, a linear operator $T : V \to V$, and two ordered bases $\mathcal{E}, \mathcal{F}$ such that

$$A = [T]_{\mathcal{E}}, \qquad B = [T]_{\mathcal{F}}.$$

If P converts coordinates from $\mathcal{E}$ to $\mathcal{F}$, so that $[v]_{\mathcal{F}} = P[v]_{\mathcal{E}}$, then

$$A = P^{-1}BP.$$

Thus similarity describes the same operation in two coordinate systems.

PROPOSITION 10.3

If $A \sim B$, the following quantities and properties are preserved:

1. **Determinant:** $\det(A) = \det(B)$.
2. **Trace:** $\operatorname{tr}(A) = \operatorname{tr}(B)$.
3. **Characteristic polynomial:** $P_A(x) = P_B(x)$.
4. **Eigenvalues:** A and B have exactly the same eigenvalues, with the same algebraic and geometric multiplicities.
5. **Rank:** $\operatorname{rank}(A) = \operatorname{rank}(B)$.
6. **Diagonalizability:** A is diagonalizable $\iff$ B is diagonalizable.

NOTE

Warning:
Equality of characteristic polynomials (and hence of eigenvalues, determinants, and traces) **does not guarantee** similarity.
A standard counterexample is $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. Both have characteristic polynomial x^2, but they are not similar: the zero matrix is similar only to itself (prove that the same is true of every scalar matrix).

EXERCISE 10.16 EXAM QUESTION

Please prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that
$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$
2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies
$$N(AB) = N(B).$$

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EXERCISE 10.16 EXAM QUESTION CONTINUED

3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\operatorname{tr}(A) = k$ that is similar only to itself.

WORKED SOLUTION

1. The claim is false. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}^2$. Appropriate bases are

$$B_{\mathbb{C}} = ((1,0),(0,1)), \qquad B_{\mathbb{R}} = ((1,0),(i,0),(0,1),(0,i)).$$

$$\dim_{\mathbb{C}} V = 2, \qquad \dim_{\mathbb{R}} V = 4 = 2\dim_{\mathbb{C}} V.$$

2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and

$$ABx = A(Bx) = A0 = 0.$$

so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives

$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$

so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.

3. True. Define

$$A = \frac{k}{n} I_n.$$

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n} I_n\right) P = \frac{k}{n} P^{-1}P = \frac{k}{n} I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

10.6 Eigenvalues and Eigenvectors

10.6.1 Motivation and Background

MOTIVATION AND GUIDING IDEA

Eigenvalues and eigenvectors help simplify complex systems. Linear operators (and matrices) generally rotate vectors and mix their coordinates. We seek nonzero vectors on which the action is multiplication by a scalar. Such vectors are **eigenvectors** , and the scalar is an **eigenvalue** . Over $\mathbb{R}$, a positive eigenvalue stretches or contracts along the same direction, a negative one also reverses direction, and a zero eigenvalue sends the vector to zero.

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Eigenvalues describe models throughout engineering. The following examples show how one algebraic equation appears in different settings:

- **Google's ranking algorithm (PageRank):** In a random browsing model, a transition matrix M sends a probability distribution to the next one. A stationary ranking distribution $\vec{r}$ satisfies
$$M\vec{r} = \vec{r}, \qquad r_i \geq 0, \qquad \sum_i r_i = 1.$$
This is an eigenvalue equation with $\lambda = 1$. The full PageRank model adds a random jump mechanism to address the link structure; the exercise below studies a simplified model.

- **Quantum mechanics:** In a finite-dimensional quantum model, a pure state is a normalized vector and a Hermitian Hamiltonian H represents energy. The eigenvalue equation
$$H\vec{\psi} = E\vec{\psi}$$
identifies states of definite energy and possible energy values. It is an application of the same linear equation; a general description of mixed states requires the additional concept of a density matrix.

- **Systems of differential equations:** If A is diagonalizable and $A = PDP^{-1}$, substituting $\vec{y} = P\vec{z}$ into $\vec{y}' = A\vec{y}$ gives
$$P\vec{z}' = AP\vec{z} = PD\vec{z} \quad \Longrightarrow \quad \vec{z}' = D\vec{z}.$$
Each component satisfies $z_i' = \lambda_i z_i$, so $z_i(t) = z_i(0)e^{\lambda_i t}$. For a continuous linear system, all solutions decay to zero if every eigenvalue has negative real part. If an eigenvalue has positive real part, some solutions grow. A zero real part requires further analysis; its sign alone does not imply decay.

- **Classifying critical points in calculus:** For a twice continuously differentiable real function, consider the symmetric Hessian at a critical point. If all its eigenvalues are positive, the point is a strict local minimum; if all are negative, it is a strict local maximum. Positive and negative eigenvalues imply a saddle point. When zero eigenvalues occur without both signs, the second derivative test may be inconclusive.

DEFINITION 10.5

Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map (a linear operator). A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a **nonzero** vector $\vec{v} \in V$ (that is, $\vec{v} \neq \vec{0}_V$) such that
$$T(\vec{v}) = \lambda\vec{v}$$
This is the **eigenvalue equation** . The vector $\vec{v}$ is an **eigenvector** of T associated with λ.

DEFINITION 10.6

Let $A \in \mathbb{F}^{n\times n}$ be a square matrix of size n.
A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there is a **nonzero** column vector $\vec{x} \in \mathbb{F}^n$ (that is, $\vec{x} \neq \vec{0}$) such that
$$A\vec{x} = \lambda\vec{x}$$
This is the **eigenvalue equation** . The vector $\vec{x}$ is an **eigenvector** of A associated with λ.

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NOTE

Over $\mathbb{R}$, the line span$\{\vec{x}\}$ is invariant. If $\lambda < 0$, direction is reversed; if $\lambda = 0$, the image is the zero vector. The algebraic concept also applies over $\mathbb{C}$, but a complex eigenvalue cannot always be interpreted as stretching a real line.

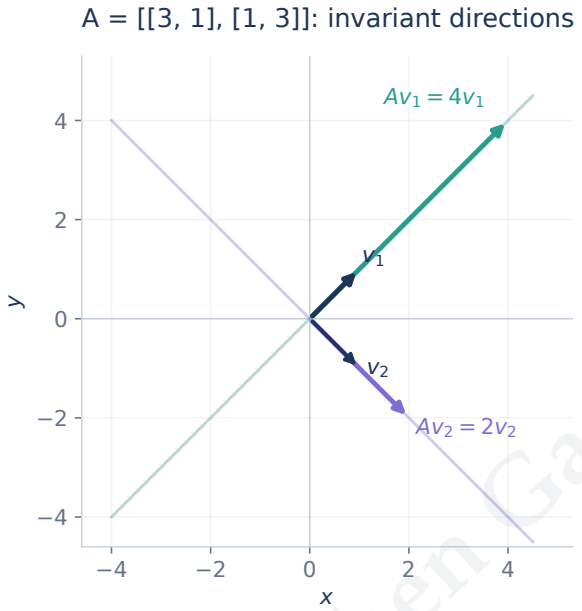
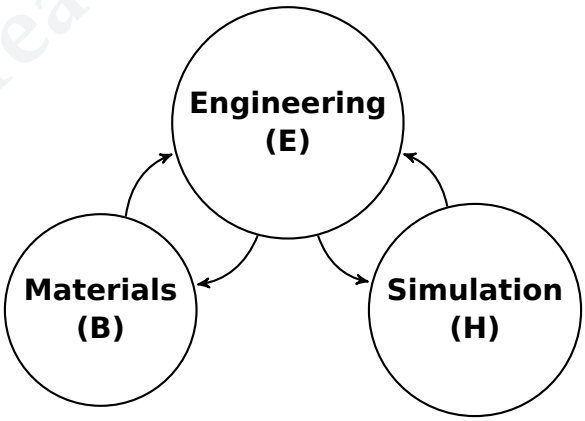


Figure 10.2. Vectors in eigendirections remain on the lines they span. A general vector may change direction because its components are stretched by different factors.

EXERCISE **10.17**

Three sites B, E, H are linked as follows. From either B or H, the next site is always E. From E, the next site is B with probability $1/3$ and H with probability $2/3$.



Please find a stationary ranking distribution whose entries sum to 1, and please compare the three rankings.

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EXERCISE 10.17 CONTINUED

WORKED SOLUTION

In the order (B, E, H), the transition matrix is

$$M = \begin{pmatrix} 0 & 1/3 & 0 \\ 1 & 0 & 1 \\ 0 & 2/3 & 0 \end{pmatrix}.$$

The equation $Mr = r$ gives

$$r_B = \tfrac{1}{3} r_E, \qquad r_H = \tfrac{2}{3} r_E, \qquad r_B + r_H = r_E.$$

Thus $1 = r_B + r_E + r_H = 2r_E$, so

$$r = (1/6, 1/2, 1/3)^T.$$

Site E has three times the rank of B and $3/2$ times the rank of H. A stationary distribution does not guarantee convergence from every start. Here

$$(0, 1, 0)^T \longmapsto (1/3, 0, 2/3)^T \longmapsto (0, 1, 0)^T,$$

so this initial distribution produces a cycle of length 2.

EXERCISE 10.18

On the space V of all real sequences, define

$$T(x_1, x_2, x_3, \ldots) = (x_1, 3x_2, 5x_3, \ldots), \qquad (Tx)_n = (2n-1)x_n.$$

Please find all eigenvalues and all eigenvectors of T.

WORKED SOLUTION

In each coordinate, the eigenvalue equation is

$$(2n-1)x_n = \lambda x_n, \qquad (2n-1-\lambda)x_n = 0.$$

If λ is not a positive odd integer, every coefficient is nonzero, so $x = 0$. For $\lambda = 2m-1$, only coordinate m can be nonzero. Therefore

$$\lambda_m = 2m-1, \qquad E_{\lambda_m} = \operatorname{span}\{e_m\}, \qquad m = 1, 2, \ldots.$$

All eigenvectors are ce_m with $c \neq 0$. For example, $Te_1 = e_1$ and $Te_2 = 3e_2$.

NOTE

The exercise concerns all sequences, with no summability or norm assumption. The unit sequences $\{e_n\}$ are not an algebraic basis of this space: finite linear combinations give only finitely supported sequences, not, for example, $(1, 1, 1, \ldots)$. An algebraically similar model can use indices $n = 0, 1, 2, \ldots$ and the rule $Ne_n = ne_n$. This resembles a number operator. The exercise alone does not imply that all physical energies are discrete or that every physical operator acts on the space of all sequences.

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EXERCISE 10.19

Let $n \geq 2$, let $V = M_n(\mathbb{C})$ be the space of square matrices, and let $p, q \in \mathbb{C}$ with $q \neq 0$. Define the linear operator $T : V \to V$ by

$$T(A) = pA + qA^T$$

Please find the eigenvalues and eigenvectors of T directly from the eigenvalue equation.

WORKED SOLUTION

Let $A \neq 0$ be an eigenvector of T with eigenvalue λ. Then

$$pA + qA^T = \lambda A \tag{10.1}$$

Isolate the transpose term:

$$qA^T = (\lambda - p)A$$

Divide by q (which is nonzero) to express A^T in terms of A:

$$A^T = \frac{\lambda - p}{q} A \tag{10.2}$$

Transpose both sides of (10.2):

$$(A^T)^T = \left(\frac{\lambda - p}{q} A\right)^T$$

Using the properties of transposition,

$$A = \frac{\lambda - p}{q} A^T$$

Substitute the expression for A^T from (10.2) into the last equation:

$$A = \frac{\lambda - p}{q} \cdot \underbrace{\left(\frac{\lambda - p}{q} A\right)}_{A^T}$$

$$A = \left(\frac{\lambda - p}{q}\right)^2 A$$

Rearrange and factor out A on the right:

$$\left[1 - \left(\frac{\lambda - p}{q}\right)^2\right] A = 0$$

Since $A \neq 0$ (an eigenvector), the scalar in brackets must vanish, so $\left(\frac{\lambda - p}{q}\right)^2 = 1$. There are two cases:

1. **First case:** $\frac{\lambda - p}{q} = 1$. Solving for λ gives the first eigenvalue:

$$\lambda_1 = p + q$$

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EXERCISE 10.19 CONTINUED

To identify its eigenvectors, substitute $\frac{\lambda-p}{q} = 1$ directly into (10.2):

$$A^T = 1 \cdot A \implies A^T = A$$

Thus the eigenvectors for λ_1 are precisely the nonzero symmetric matrices.

2. **Second case:** $\frac{\lambda-p}{q} = -1$. Solving for λ gives the second eigenvalue:

$$\lambda_2 = p - q$$

Substitute $\frac{\lambda-p}{q} = -1$ into (10.2):

$$A^T = -1 \cdot A \implies A^T = -A$$

Thus the eigenvectors for λ_2 are precisely the nonzero skew-symmetric matrices.

10.7 An Introduction to Diagonalization

DEFINITION 10.7

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible matrix $P \in M_n(\mathbb{F})$ and a diagonal matrix $D \in M_n(\mathbb{F})$ such that $A = PDP^{-1}$. If p_j is column j of P and d_j is diagonal entry j of D, then $AP = PD$ means precisely

$$Ap_j = d_j p_j \qquad (1 \le j \le n).$$

Invertibility of P means its columns form a basis. Thus diagonalization amounts to choosing a basis of eigenvectors. In those coordinates, each component is multiplied by its corresponding scalar.

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EXAMPLE 10.3

Over $\mathbb{R}$, consider

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have

$$A\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The vectors are independent because their column matrix has determinant -5. Hence

$$P = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}, \quad P^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The inverse can be checked using $P^2 = 5I_2$. Also,

$$AP = \begin{pmatrix} 10 & 0 \\ 5 & 0 \end{pmatrix} = PD, \qquad PDP^{-1} = \begin{pmatrix} 2 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = A.$$

Each vector has a unique decomposition $x = \alpha(2,1)^T + \beta(1,-2)^T$, and

$$Ax = 5\alpha(2,1)^T.$$

Thus the action in the new coordinates is $(\alpha, \beta)^T \mapsto (5\alpha, 0)^T$.

EXAMPLE 10.4

Over $\mathbb{R}$, the matrix

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

has no basis of eigenvectors. Its eigenvalue equation is

$$(J - \lambda I)\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} (2-\lambda)x + y \\ (2-\lambda)y \end{pmatrix} = 0.$$

If $\lambda \neq 2$, the second equation gives $y = 0$, and the first then gives $x = 0$. Thus the only eigenvalue is 2. For that value, $y = 0$, so all eigenvectors lie on $\text{span}\{(1,0)^T\}$. Two independent vectors cannot be chosen from that line. Hence J is not diagonalizable.

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EXERCISE 10.20 CHALLENGE

Let $\dim V = n$ over a field F, and let $T : V \to V$ be linear with $T^2 = 0$. Put $r = \operatorname{rank} T$. Please prove that $2r \le n$ and that some basis gives the matrix

$$\begin{pmatrix} 0 & I_r & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three row and column block sizes are $r, r, n - 2r$. Blocks of size zero are omitted.

WORKED SOLUTION

The identity $T^2 = 0$ gives $\operatorname{Im} T \subseteq \ker T$. Rank-nullity yields $r \le n - r$, so $2r \le n$. Choose a basis $u_1, \dots, u_r$ of the image and extend it to a basis $u_1, \dots, u_r, z_1, \dots, z_{n-2r}$ of the kernel. For each i, choose w_i with $T(w_i) = u_i$. The list

$$\mathcal{B} = (u_1, \dots, u_r, w_1, \dots, w_r, z_1, \dots, z_{n-2r})$$

is independent. Applying T to a zero linear combination shows that all coefficients of the w_i vanish, because the u_i are independent. The remaining combination is in a basis of the kernel, so all its coefficients vanish too. The list has n vectors and is therefore a basis. In this basis $T(u_i) = T(z_j) = 0$ and $T(w_i) = u_i$. Writing these images as columns gives the displayed matrix. When $r = 0$, $T = 0$ and every basis works.

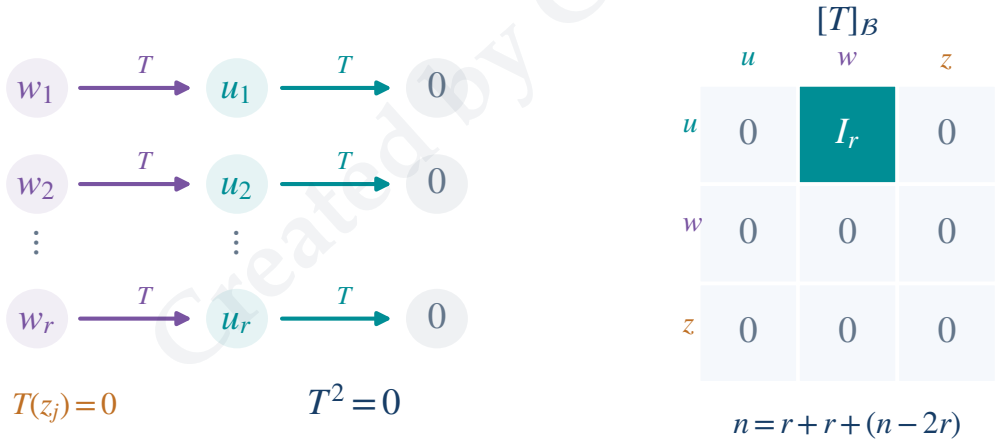


Figure 10.3. The schematic shows the case $r \ge 3$. Each w_i maps to u_i, and each u_i maps to zero. The vectors z_j complete $u_1, \dots, u_r$ to a basis of the kernel. The basis groups are ordered u, w, z.

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EXERCISE 10.21 CHALLENGE

Let $A \in M_n(F)$ have rank 1, where $n \geq 2$, and put $\alpha = \operatorname{tr} A$, the sum of the diagonal entries. Please prove that A is similar to

$$\begin{cases} \operatorname{diag}(\alpha, 0, \ldots, 0), & \alpha \neq 0, \\ E_{12}, & \alpha = 0, \end{cases}$$

where E_{12} has a single 1 in position $(1, 2)$. Please deduce that two rank-one matrices of the same size and trace are similar.

WORKED SOLUTION

Write $A = uv^T$ with $u, v \neq 0$ and define $\varphi(x) = v^T x$. Then

$$Ax = \varphi(x)u, \qquad \operatorname{tr} A = \sum_i u_i v_i = \varphi(u) = \alpha.$$

Since $\varphi \neq 0$, its kernel has dimension $n-1$. If $\alpha \neq 0$, then $u \notin \ker \varphi$. A basis $\mathcal{B} = (u, z_2, \ldots, z_n)$ obtained by adjoining u to a basis of the kernel satisfies $Au = \alpha u$ and $Az_j = 0$. Its representing matrix is $\operatorname{diag}(\alpha, 0, \ldots, 0)$. If $\alpha = 0$, then $u \in \ker \varphi$. Choose w with $\varphi(w) = 1$ and extend u to a basis $(u, z_3, \ldots, z_n)$ of the kernel. The list $\mathcal{B} = (u, w, z_3, \ldots, z_n)$ is a basis: applying φ to a zero combination forces the coefficient of w to vanish; the remaining coefficients vanish by independence in the kernel. Now $Au = Az_j = 0$ and $Aw = u$, giving the matrix E_{12}. Two rank-one matrices with the same size and trace are therefore similar to the same displayed matrix, hence to one another.

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